Ayoung Park

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WORK EXPERIENCE

Senior Manager Jan 2026 - present
Supervisory Strategy Research Team, Supervision Innovation Department, Financial Supervisory Service, Seoul, Republic of Korea

- Conducted research on blockchain-based financial networks, regulatory frameworks, and supervisory innovation (including AML and crypto-assets).

Senior Manager Mar 2024 - Jan 2026
Future Finance Research Team, Financial Market Stabilization Department, Financial Supervisory Service, Seoul, Republic of Korea

- Conducted research on Financial Market Analysis related with the financial stability, regulation, and innovation.

Research Assistant Professor July 2021 - Feb 2024
The Institute of Knowledge Management, Division of Business Administration, Chosun University, Gwangju, Republic of Korea

- Conducted research on Financial Market Analysis via Connectedness and Deep Learning with the 5-years funding from National Research Foundation(NRF) of Korea

Visiting Scholar Jan 2023 - Feb 2023
Department of Anesthesiology, University of Michigan Medical School, Ann Arbor, MI, United States

- Research on the prediction of the speed of collapse and recovery in the 2007-2009 global financial crisis on international stock markets via complex network theory.

Lecturer Mar 2021 - June 2021
Division of Business Administration, Chosun University, Gwangju, Republic of Korea

- Played a pivotal role in the conceptualization and creation of the *Introduction to Fintech* course, ensuring its relevance and applicability in the current financial landscape.

Visiting Scholar Mar 2019 - Feb 2020
Department of Anesthesiology, University of Michigan Medical School, Ann Arbor, MI, United States

- Research on the general mechanism of response and recovery between anesthesia on the brain and the impact of the 2007-2009 global financial crisis on international stock markets.

Visiting Scholar Jan 2019 - Feb 2019
Department of Physics, Pohange University of Science and Technology (POSTECH), Pohang, Republic of Korea

- Research on the non-linear dynamics and complexity based on brain and financial market.

Visiting Scholar Jan 2017 - Feb 2017
Division of Business Administration, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, United States
- Research on the role of board diversity and behavior finance.

Education

Ph.D. in Finance Feb 2021
Chosun University, Gwangju, Republic of Korea
- Dissertation title: Essays on Financial Connectedness, Systemic Risk, and Performance.
- Excellent Research Award

M.S. in Finance Aug 2016
Chosun University, Gwangju, Republic of Korea
- Thesis title: The Empirical Study on Measuring Systemic Risk using Variance Decomposition and Debtrank method

B.Sc. in Mathematics and B.A. in Business Administration Feb 2015
Chosun University, Gwangju, Republic of Korea
- Summa Cum Laude

PROFESSIONAL TRAINING

Anti-Money Laundering(AML) Specialist Program, Hanyang University, Dec 2025 – Focus on AML regulation, crypto-assets, and financial surveillance

RESEARCH INTERESTS

Computational Finance, Risk Management, Financial Intermediation, Empirical Asset Pricing, Corporate Governance, Complex System, Graph Neural Network, Natural Language Processing

RESEARCH FUNDING

Title: *Financial Market Analysis via Connectedness and Deep Learning.*
- **Objective:** Investigating the connectedness among financial institutions and boards of directors from international countries. The focus lies on risk propagation, knowledge spillover, and its predictive potential for technological innovation.
- **Source:** National Research Foundation of Korea (NRF).
- **Amount:** A grant of 200 million KRW.
- **Duration:** July 2021 - Mar 2024.

WORKING PAPERS

- **Global Banking Networks and Loan Pricing**, Job Market Paper
  - **Best Paper Award, Semi-finalist in Financial Intermediation & Markets at Financial Management Association (FMA) 2023**
  - **Best Student Paper Award** at Fall conference of Korea Academy of Complexity Studies 2018 with a prize of 200,000 KRW (approx. 181 USD)
  - **Computing in Economics and Finance (CEF) 2023**
  - **Workshop of Asia- Pacific Applied Economics Association, 2023**
- **Stablecoin Prices as Shadow FX Quotes: Dollar News and the Korean Crypto Premium**
  - **Financial Management Association Annual Meeting, 2026**
- **Model Disagreement and Forecast Triage under Limited Capacity: Evidence from New York Fed Policy-Rate Paths**

- **Investor Sentiment and Stock Return Predictability: The Influence of Online Stock Discussion Boards**, (with Danbi Park, Gabjin Oh)

JOURNAL ARTICLES

See also my google scholar page.

- Uncheol Lee, Hyoungkyu Kim, Minkyung Kim, Gabjin Oh, Pangyu Joo, Ayoung Park, Jamie Sleigh, Katie Warnaby, Iren Tracey, George Mashour, Proximity to Explosive Synchronization Determines Network Collapse and Recovery Trajectories in Neural and Economic Crises, *PNAS*, (SCI, impact factor: 9.1, Ranked 14/135 in Multidisciplinary Sciences), Sep 2025.
- Ayoung Park and Gabjin Oh, The Contagion Effect of Heterogeneous Investor Groups, *Plos One* (SCI, impact factor: 3.752, Ranked 29/134 in Multidisciplinary Sciences), Oct 2023.
- Ayoung Park and Gabjin Oh, The Core of the Board Networks and Firm Value, *Frontiers in Physics* (SCI, impact factor: 3.718, Ranked 42/111 in Physics, Multidisciplinary), Feb 2023.
- Ayoung Park and Gabjin Oh, Economic System Research by Using Statistical Physics, *Physics & High technology* (Highlighted Article), Aug 2022.
- Jinjoo Yoon, Ayoung Park, and Gabjin Oh, Grouping Analysis of Social Media in Korean Financial Market, *New Physics: Sae Mulli* (SCOPUS), Feb 2022.
- Ayoung Park and Gabjin Oh, Lending Diversification and Interconnectedness of Syndicated Loan Market, *Frontiers in Physics* (SCI, impact factor: 3.56, Ranked 30/110 in Physics, Multidisciplinary), Jan 2021.
- Jie Deng, A-young Park, and Gabjin Oh, The Relationship between Cash Flows and Investment according to Economic Status, *Journal of the Korean Physical Society* (SCI, impact factor: 0.649, Ranked 87/111 in Physics, Multidisciplinary), Jan 2020.
- Ayoung Park and Gabjin Oh, Systemic risk and Expected return, *New Physics: Sae Mulli* (SCOPUS), June. 2018.
- Ayoung Park and Gabjin Oh, An Analysis of the Relationship between Market Risk and Information Flows among Trading Volume of the Industry Sector in the Korean Stock Market, *New Physics: Sae Mulli* (SCOPUS), May. 2017.
- Gabjin Oh, Ho-Yong Kim, and Ayoung Park, Analysis of technological innovation based on citation information, *Quality and Quantity* (SSCI, impact factor: 1.25, Ranked 42/96 in Social Sciences, Interdisciplinary), Dec. 2016.
- A Young Park, Ho-Yong Kim, and Gabjin OH, An Empirical Study on Measuring Systemic Risk Based on Information Flows using Variance Decomposition and DebtRank, *The Korean Operations Research and Management Science Society*, (KCI), Nov. 2015.

WORK IN PROGRESS

- **Forecast Compression and Borrower-Size Gradients in Syndicated Lending**
- **Holdings Embeddings and Subsequent Illiquidity**
- **Missing Carbon and Model Risk**
- **Predictable, Not Deployable: Economic Networks and the Transportability of Housing Forecasts**

TEACHING EXPERIENCE

Introduction to Fintech, undergraduate, Chosun University (Spring 2021-2023)
Security Investment Practice, undergraduate, Chosun University (Spring 2023)
Management Statistics, undergraduate, Chosun University (Fall 2021, 2023)
Management and Economics, undergraduate, Chosun University (Fall 2021-2023)

SPECIAL ACHIEVEMENTS

Awards

- *Best Paper Award*, Semi-finalist in Financial Intermediation at Financial Management Association (FMA), 2023
- *Ph.D. Excellent Research Award*, Chosun University, 2021
- *Best Research Award* for "Measuring Information Asymmetry of Interbank Network from Syndicated Loan Market", Fall Conference of Korea Academy of Complexity Studies, 2018
- *Best Research Award* for "Diversity in the Board Network Impact on Firm Performance", Fall Conference of Korea Academy of Complexity Studies, 2017
- *Encouragement Award* for "Analysis of Financial Stability Based on Investor Trading Behavior", 4th Conference of Social Sciences Korea Networking, 2017
- *Fellowship*, Brain Korea 21+, Korean Government, 2015-2020
- *Summa Cum Laude*, Department of Mathematics, Chosun University, 2014
- *Scholarship*, Catholic Archdio47cese of Gwangju, 2014
- *Scholarship*, KT&G Scholarship Foundation, 2013-2014
- *Full-Tuition Scholarship*, Korea Student Aid Foundation (KOSAF), 2011-2014

Industry Experience

- *Researchers*, Technological Innovation and Data Construction for Financial/Economic Digital Twins in Companies and Industries at ETRI(Electronics and Telecommunications Research Institute), 2022- 2023

Volunteer Experience

- *Global Volunteer*, Happy Move funded Hyundai and Habitat: Participated in community house-building in the Philippines and intercultural exchange, Feb 2014
- *KT&G 7st SangSang Univ*, University Student Planning Group, 2013
- *Education*, Elementary school in Catholic, 2011-2014.

Invited Lectures

- Global Banking Network and Syndicated Loan Prices, at workshop of Asia-Pacific Applied Economics Association, Xian Jiaotong University, July. 2023
- Network Analysis in the Financial System, Invited Talk at conference of Korea Academy of Complexity Studies, Sookmyung Women's University, Dec. 2022; at the Ajou University, Oct. 2022; at the Pukyong National University, Aug. 2022
- From Research to Regulatory Affairs, Invited Talk at Inha University, Nov. 2025

PROFESSIONAL INTERVIEWS

- Invited Interview in National Research Foundation of Korea(NRF), Oct 2021 (based on the high score of research project)

CONFERENCE CONTRIBUTIONS

- When the Onshore FX Market Closes: Information in KRW Stablecoin Prices for the Won-Dollar Exchange Rate, *2026 FMA*
- Global Banking Network and Syndicated Loan Prices, *Sydney Banking and Finance Conference (SBFC) 2025, 2023 Financial Management Association (FMA), 29th International Conference Computing in Economics and Finance (CEF), 2023 Fall Conference of Korean Finance Association, 15th Asia Pacific Physics Conference, 24th Interdisciplinary KASBA conference, Networks 2021: A Joint Sunbelt and NetSci Conference, 2019 The Network Science Society*
- A Network Approach to Optimal Portfolio: Application to the Korean Stock Market, *29th International Conference Computing in Economics and Finance (CEF)*
- Measuring Similarity between trend Behaviors of Multivariate Time Series, *2023 European Financial Management Association (EFMA), 28th International Conference Computing in Economics and Finance (CEF)*
- Evolution of Social media activity-financial market Based on the complex network method, *28th International Conference on Statistical Physics*
- The Effect of Knowledge Diversity of Board Networks on Corporate Value, *27th International Conference Computing in Economics and Finance (CEF), 2018 Econophysics Colloquium, 2018 Asia Pacific Econophysics Conference, 2017 Fall conference of Korea Academy of Complexity Studies*
- Analysis of Mutual Fund Simiarlity Network in Korea Equity Market, *2018 Asia Pacific Econophysics Conference, The 13th INFINITI Conference on International Finance*
- Financial Stability Based on Information Flows among Investor Trading Behavior, *25th International Conference Computing in Economics and Finance (CEF), 2018 Asia Pacific Econophysics Conference, NetworkX-2018: International School and Conference on Network Science, 2017 International Conference on SigmaPhi, 2017 4th conference of Social Sciences Korea Networking (SSK), 2016 Japanese Association of Financial Econometrics and Engineering (JAFEE)*
  - Encouragement student paper award at the 4th conference of Social Sciences Korea Networking (SSK).
- Usefulness of the Fama and French three-factor model in Shanghai A-share market, *2016 International School and Conference on Network Science, 2016 Fall conference of Korea Academy of Complexity Studies*

PROFESSIONAL SERVICE

- **Journal Referee:** Plos One, Journal of the Korea Society Industrial Information System, Humanities and Social Sciences Communications
- **Conference Discussant:** Fall conference of Korean Finance Association 2022; European Financial Management Association (EFMA) 2023; Financial Management Association (FMA) 2023, Asia Financial Management Association (Asia FMA) 2024; APAD 2025; Sydney Banking and Financial Stability Conference (SBFC) 2025; FMA 2026
- **Conference Review Committee:** Financial Management Association (FMA) 2023, Sydney Banking and Financial Stability Conference (SBFC), 2026
- **Conference Session Chair:** Computing in Economics and Finance (CEF) 2022, 2023; Financial Management Association (FMA) 2023; Asia Financial Management Association (Asia FMA) 2024; FMA 2026

- **Professional Membership:** Financial Management Association (FMA), Korean Security Association, Korea-America Finance Association(KAFA)

SKILLS Python, Stata, Matlab, C++, Latex, Gephi

LANGUAGES English (Fluent), Korean (Native)

REFERENCE

- **Gabjin Oh, Ph.D.**, Professor, Division of Business Administration, Chosun University, Gwangju, Republic of Korea, *phecogjoh@chosun.ac.kr*
- **Uncheol Lee, Ph.D.**, Associate Professor, Department of Anesthesiology, Associate Director, Center for the Study of Complex Systems, University of Michigan, Ann Arbor, MI, United States, *uclee@med.umich.edu*
- **Woo-Sung Jung, Ph.D.**, Professor, Department of Industrial and Management Engineering/Physics, Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea, *wsjung@postech.ac.kr*
- **O-Kyu Kwon, Ph.D.**, Senior Researcher, NIMS-National Institute for Mathematical Sciences, South Korea, *okyu.kwon@gmail.com*
- **Hyeng Keun Koo, Ph.D.**, Professor, Department of Financial Engineering, Ajou University, Suwon, Republic of Korea, *hkoo@ajou.ac.kr*